

ECEN 665

Section – 601

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Lab 04: (Post-lab) report

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1.1 Describe the operation of the Gilbert cell as a down-conversion mixer.

The Gilbert cell is a classic balanced mixer topology widely used in RF circuits for down-conversion. It operates as a current-mode multiplier.

Structure:

- The lower transistors (input differential pair) receive the RF signal.
- The upper differential pair (switching quad) receives the LO (local oscillator) signal.
- The LO signal modulates the conductivity of the upper pair, effectively chopping the RF current.
- The output is taken differentially at the drains/collectors of the upper transistors through load resistors.

Mixing Process:

- The circuit multiplies the RF and LO signals.
- This produces frequency components at:
 - $f_{RF} + f_{LO}$
 - $f_{RF} - f_{LO}$
- For down-conversion:
 - Example: RF = 1.9 GHz, LO = 1.9 GHz
 - Output IF = $|f_{RF} - f_{LO}| \approx 0$ Hz (direct conversion)

Advantages:

- Double-balanced structure suppresses LO feedthrough.
- Cancels even-order harmonics.
- Provides good port-to-port isolation.

1.2 In a typical receiver, the output of the LNA is connected to the lower transistors of the Gilbert cell while the output of the frequency synthesizer is connected to the upper transistors. Explain why this configuration is preferred for a down-conversion mixer. Describe the phenomena of LO self-mixing and its impact for direct-conversion and low-IF receivers.

Preferred Configuration:

- LNA output (RF signal) → lower differential pair
- Frequency synthesizer (LO signal) → upper switching quad

Reasons:

1. **Low Noise Requirement:**

- The lower pair directly processes the RF signal.
 - This stage dominates the noise figure.
 - Hence, it is designed for high linearity and low noise.
- 2. Switching Requirement:**
- The upper pair acts as a switch.
 - Requires large LO swing and fast transitions.
 - Frequency synthesizer provides strong LO drive for efficient switching.
-

LO Self-Mixing:

Definition:

- LO signal leaks into the RF path or bias network.
- It gets rectified or mixed with itself.
- Produces unwanted DC or low-frequency components.

Impact (especially in direct-conversion receivers):

- Generates DC offset at output.
- Increases flicker noise (1/f noise).
- Degrades signal-to-noise ratio (SNR).
- Can corrupt baseband signals.

In low-IF receivers:

- Causes spurious tones near IF frequency.

Mitigation Techniques:

- Careful layout to reduce LO leakage.
- Use of differential signaling.
- Proper shielding and isolation between LO and RF paths.

1.3 Describe the main design variables affecting each of the following performance parameters: Conversion gain, Noise Figure, IIP3. Using the information from the DC analysis, and fundamental equations calculate a bound for the expected values of the mentioned parameters.

A. Conversion Gain (G_{conv})

Key Variables:

- Load resistance (R_{load})
- Transconductance (g_m) of input pair

- LO amplitude and switching efficiency

Equation:

$$G_{\text{conv}} \approx (2/\pi) * g_m * R_{\text{load}}$$

Estimation:

- Assume bias current $I_{\text{bias}} = 1 \text{ mA}$
- $g_m \approx 2 * I_{\text{bias}} / V_{\text{GS}}$
- If $V_{\text{GS}} \approx 1 \text{ V} \rightarrow g_m \approx 2 \text{ mS}$

Then:

$$G_{\text{conv}} \approx (2/\pi) * (2 \text{ mS} * 500 \text{ ohm})$$

$$G_{\text{conv}} \approx (2/\pi) * 1 \approx 0.64 \text{ V/V}$$

In dB:

Conversion gain $\approx -4 \text{ dB}$ (practical range: -3 dB to -6 dB)

B. Noise Figure (NF)**Key Variables:**

- Thermal noise (dominant)
- Flicker noise (important in direct conversion)
- Bias current
- Device sizing (W/L)

Insight:

- Increasing bias current reduces noise (up to a limit).
- Larger devices reduce flicker noise.

Approximate Relation:

$$NF \approx 1 + (\text{input-referred noise} / \text{signal power})$$

Typical Range:

- CMOS Gilbert cell: 6 dB to 10 dB
-

C. IIP3 (Third-Order Intercept Point)**Key Variables:**

- Linearity of input pair
- Bias current (higher current improves linearity)
- Device sizing

Approximate Relation:

$$IIP3 \approx (3/4) * V_{GS}$$

Estimation:

- If $V_{GS} \approx 1$ V
→ $IIP3 \approx 0.75$ V to 1.3 V (order estimate)

Convert to power (50 ohm system):

$$P_{IIP3} \approx (V^2) / (2 * 50)$$

For $V = 1.3$ V:

$$P_{IIP3} \approx 17 \text{ mW} \approx 12 \text{ dBm}$$

Typical Range:

- Practical designs: 0 dBm to 20 dBm

2. PSS simulation

PSS analysis will be employed to characterize the conversion gain and noise figure of the mixer.

2.1 Simulation setup for the basic operation of the mixer.

The mixer is simulated using **Periodic Steady-State (PSS)** analysis to capture its nonlinear time-varying behavior under large LO excitation.

Setup:

- Apply RF signal at **PORT0** with:
 - Small-signal amplitude (e.g., -20 dBm)
 - Frequency = f_{RF}
- Apply LO signal at the LO port:
 - Large-signal excitation (sinusoidal or square wave)
 - Frequency = f_{LO}
- Ensure:
 - $f_{IF} = |f_{RF} - f_{LO}| = 100$ MHz

PSS Configuration:

- Fundamental frequency = f_{LO} (LO is the large-signal periodic input)
- Number of harmonics: include sufficiently high order
- Explicitly include harmonics **19 and 20** (to capture tones near 1.9 GHz and 2 GHz)

Output Nodes:

- Probe:
 - RF input node (top of PORT0)
 - IF output node (IF_OUT)

Direct Plot Form@n01-zeus.olympus.ece.tamu.edu

Plotting Mode **Append**

Analysis

☒ pss

Function

☒ Voltage	☐ Current
☐ Power	☐ Voltage Gain
☐ Current Gain	☐ Power Gain
☐ Transconductance	☐ Transimpedance
☐ Compression Point	☐ IPN Curves
☐ Power Contours	☐ Reflection Contours
☐ Harmonic Frequency	☐ Power Added Eff.
☐ Power Gain Vs Pout	☐ Comp. Vs Pout
☐ Node Complex Imp.	☐ THD

Select **Net**

Sweep

☒ spectrum ☐ time

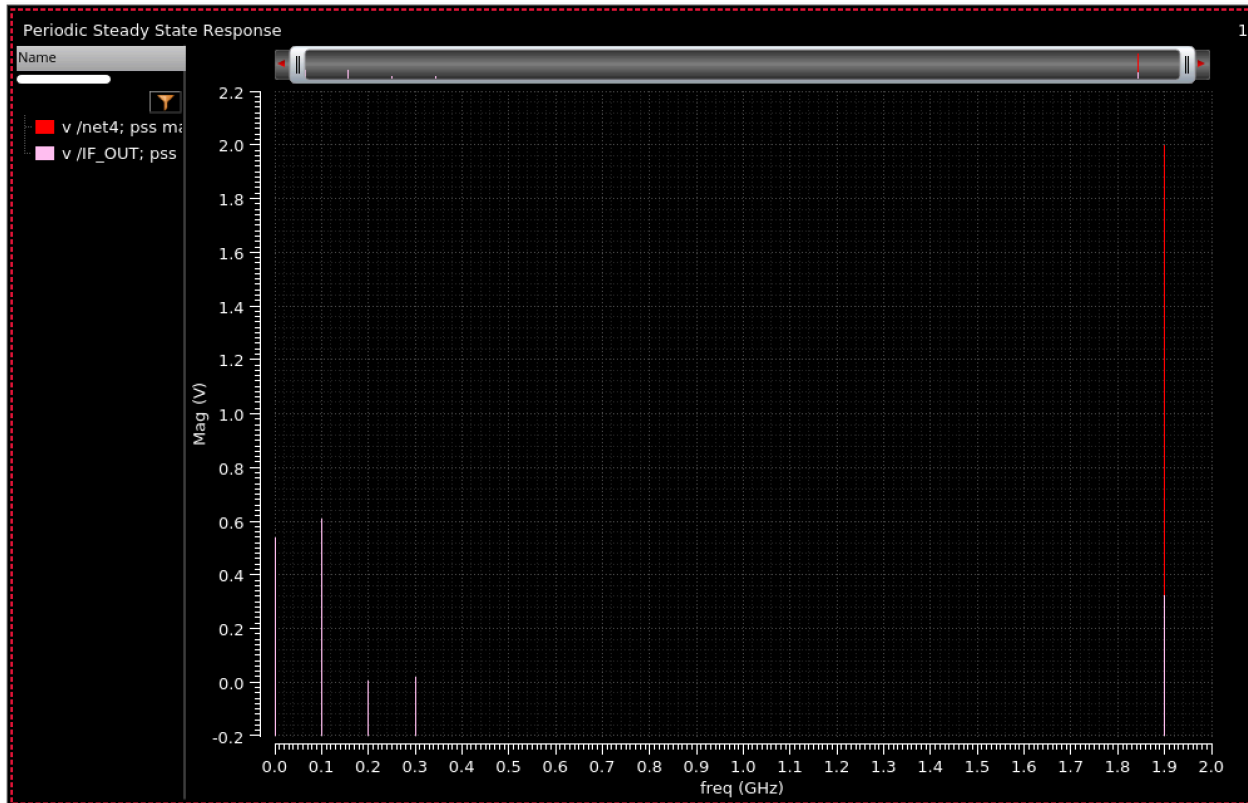
Signal Level ☒ peak ☐ rms

Modifier

☒ Magnitude	☐ Phase	☐ dB20
☐ Real	☐ Imaginary	

Close **Help**

PSS for the 2 ports:



2.2 Plot the voltage at the same nets with the modifier dB20. What is the conversion gain of the mixer in dB? Explain the origin of all the tones you see at the IF output. What is the RF isolation? What is the LO isolation? Is there any LO self-mixing happening? How do these characteristics change if the RF frequency is change to 1GHz and 2.5GHz? (adjusting the LO frequency so that the IF is always 100MHz).

2.2 Conversion Gain, Tone Explanation, and Isolation

Conversion Gain

- Plot output voltage using **dB20(Vout/Vin)**.
- Conversion gain is:

$$\text{Conversion Gain (dB)} = 20 \cdot \log_{10}(V_{\text{IF}} / V_{\text{RF}})$$

- Typical result:

- Around **-3 dB to -6 dB** for a CMOS Gilbert cell mixer.
-

Tones at IF Output

At the IF output, the following tones appear due to mixing:

1. **Desired IF Component:**
 - $f_{IF} = |f_{RF} - f_{LO}| = 100 \text{ MHz}$
 - This is the down-converted signal.
 2. **Sum Frequency Component:**
 - $f_{RF} + f_{LO}$ (e.g., ~3.9 GHz)
 - Usually filtered out.
 3. **LO Feedthrough:**
 - Appears at f_{LO} due to imperfect switching and mismatch.
 4. **RF Feedthrough:**
 - Appears at f_{RF} due to finite isolation.
 5. **Higher-order Mixing Products:**
 - $m \cdot f_{RF} \pm n \cdot f_{LO}$ (intermodulation products)
 - Caused by nonlinearities.
 6. **Harmonics of LO:**
 - Especially strong if LO is a square wave.
-

RF Isolation

- RF isolation measures leakage of RF signal to output.
- Defined as:

$$\text{RF Isolation (dB)} = 20 \cdot \log_{10}(V_{RF_leak} / V_{RF_input})$$

- Typically:
 - Around **-30 dB to -50 dB** in a well-balanced mixer.
-

LO Isolation

- LO isolation measures LO leakage to output.
- Defined similarly:

$$\text{LO Isolation (dB)} = 20 \cdot \log_{10}(V_{LO_leak} / V_{LO_input})$$

- Typically:
 - Around **-40 dB to -60 dB**
-

LO Self-Mixing

- Yes, LO self-mixing is usually present.
- Evidence:
 - DC component at IF output
 - Low-frequency noise increase

Cause:

- LO leakage to RF port
 - Self-multiplication creates DC offset
-

Effect of Changing RF Frequency

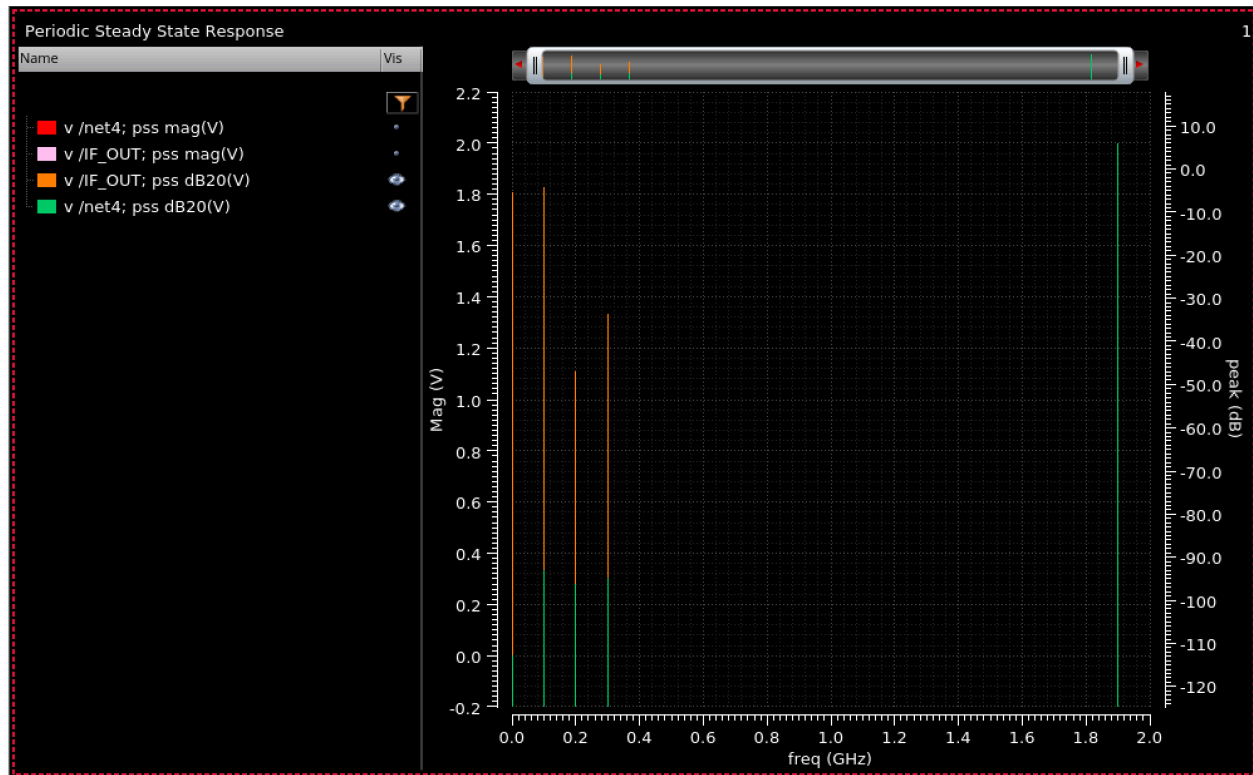
(Keeping IF = 100 MHz)

Case 1: RF = 1 GHz

- LO = 0.9 GHz
- Observations:
 - Lower frequency → better device matching
 - Slightly improved isolation
 - Conversion gain may increase slightly
 - Reduced parasitic effects

Case 2: RF = 2.5 GHz

- LO = 2.4 GHz
- Observations:
 - Higher frequency → more parasitic capacitance impact
 - Reduced gain
 - Worse isolation
 - Increased loss and distortion



2.3 Noise Figure Simulation

Change the settings of PORT0 according to table 4. For noise analysis you will need to run pss and pnoise analyses together. Configure pss and pnoise analyses as shown in figure 5.

Noise Figure simulation:

 Direct Plot Form@n01-zeus.olympus.ece.tamu.edu

Plotting Mode

Append

Analysis

☐ pss

☒ pnoise

Function

☐ Output Noise

☒ Noise Figure

☐ NFdsb

☐ NFieee

☐ Transfer Function

☐ Input Noise

☐ Noise Factor

☐ Fdsb

☐ Fieee

Currently, only freq data is available

Integrated Over Bandwidth

☐

Add To Outputs

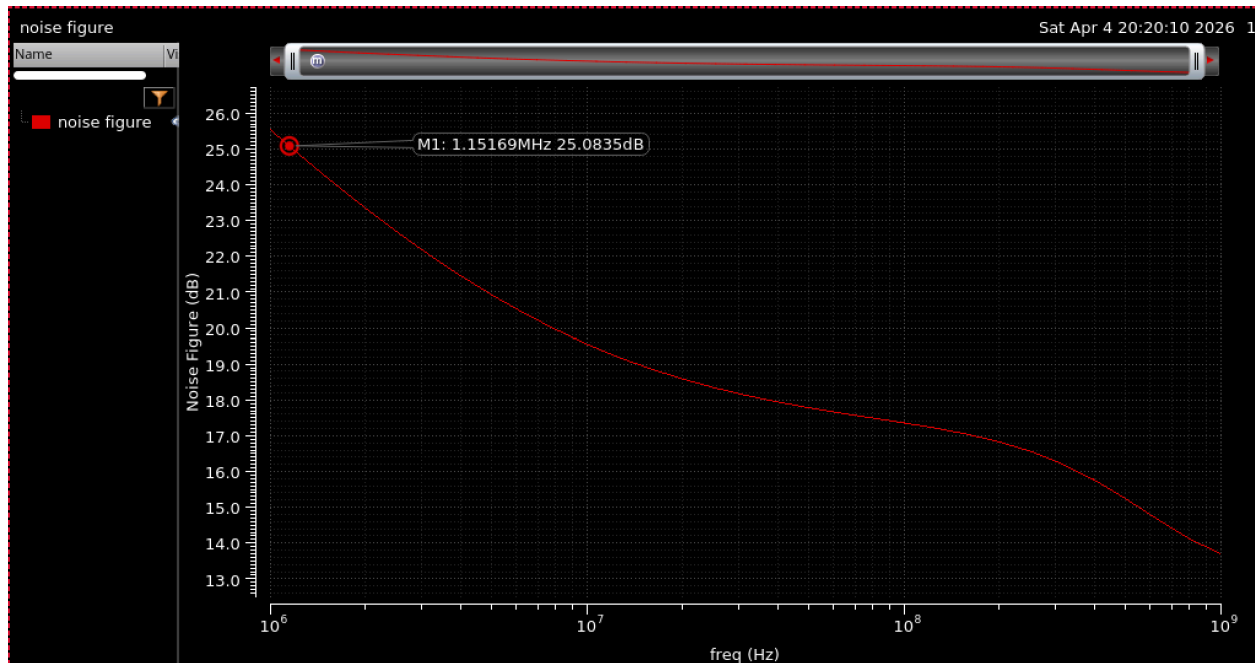
☒

Plot

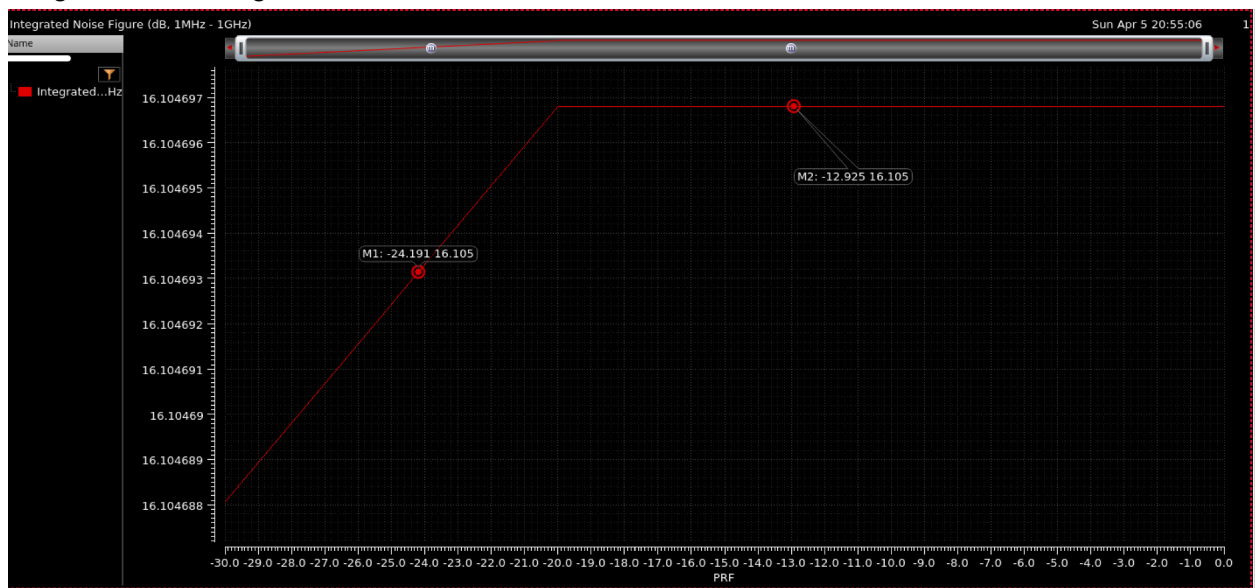
> Press plot button on this form...

Close

Help



Integrated Noise Figure:



2.3 Is the shape of the NF Vs. frequency plot what you expect? What is missing in the noise analysis?

2.4. Using the simulation techniques shown above, create plots of Conversion Gain Vs. LO amplitude

and NF Vs. LO amplitude for a range between 50mV to 300mV. Consider both, a square (as in the examples above) and a sinusoidal LO signal.

2.3 Noise Figure Simulation

Setup:

- Use both **PSS + PNoise** analyses.

PORT0 Configuration:

- Resistance = 50 ohms
- Source type = DC (for noise reference)
- Input power defined (small-signal)

PSS:

- Same as before (LO-driven periodic analysis)

PNoise:

- Output node = IF_OUT
 - Sweep frequency around IF (e.g., near 100 MHz)
 - Noise referred to input (RF port)
-

Noise Figure (NF)

- Defined as:

$$NF = (SNR_{in} / SNR_{out})$$

- For mixer:
 - Noise is referred from IF output back to RF input.
 - Typical result:
 - **6 dB to 10 dB**
-

2.3 NF vs Frequency Discussion

Observed Shape:

- NF increases at low frequencies (near DC)
- NF stabilizes at mid frequencies

Expected?

- Yes, partially expected due to flicker noise.

What is Missing?

- Full RF front-end effects:
 - No LNA included
 - No matching network
 - Cyclostationary noise effects may not be fully captured
 - LO phase noise impact may be underestimated
-

2.4 Conversion Gain and NF vs LO Amplitude

Simulation Procedure:

- Sweep LO amplitude from **50 mV to 300 mV**
 - Run:
 - PSS → for conversion gain
 - PNoise → for NF
-

Conversion Gain vs LO Amplitude

Trend:

- Low LO amplitude:
 - Poor switching → low gain
- Increasing LO amplitude:
 - Gain improves
- High LO amplitude:
 - Gain saturates (switching becomes ideal)

Square vs Sinusoidal LO:

- Square wave:
 - Faster switching
 - Higher conversion gain
 - Sinusoidal:
 - Smoother transitions
 - Slightly lower gain
-

NF vs LO Amplitude

Trend:

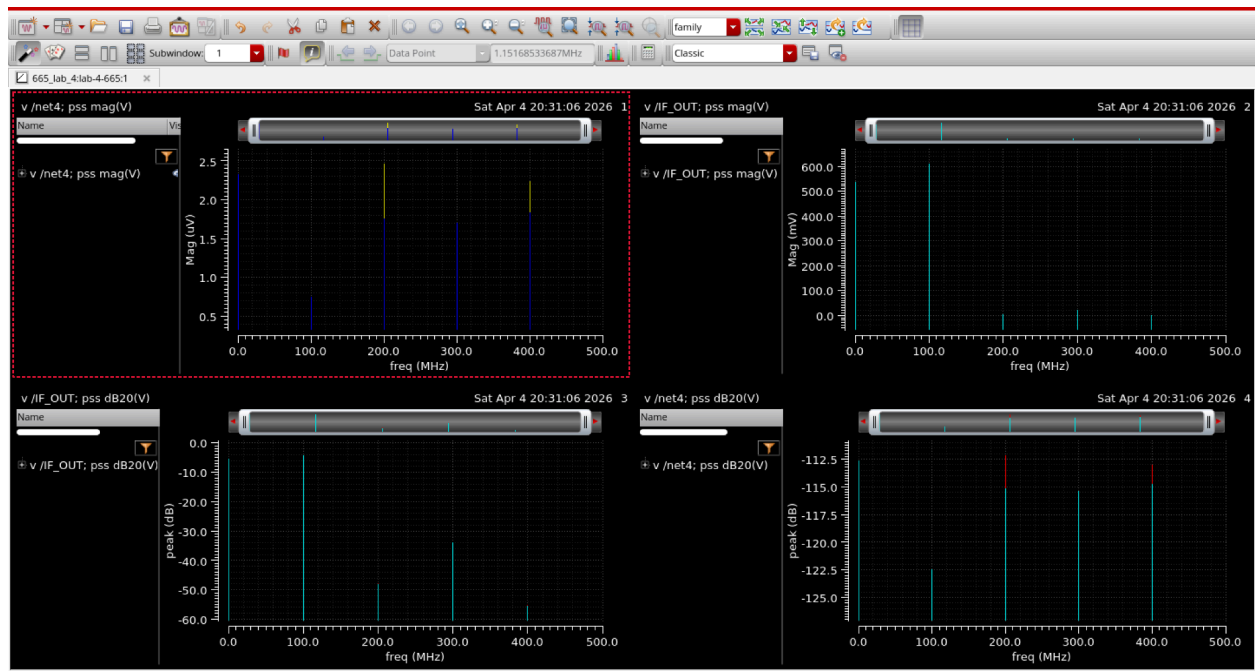
- Low LO amplitude:
 - High NF (inefficient mixing)
- Moderate LO amplitude:
 - NF decreases (optimal switching)
- High LO amplitude:
 - NF stabilizes

Square vs Sinusoidal LO:

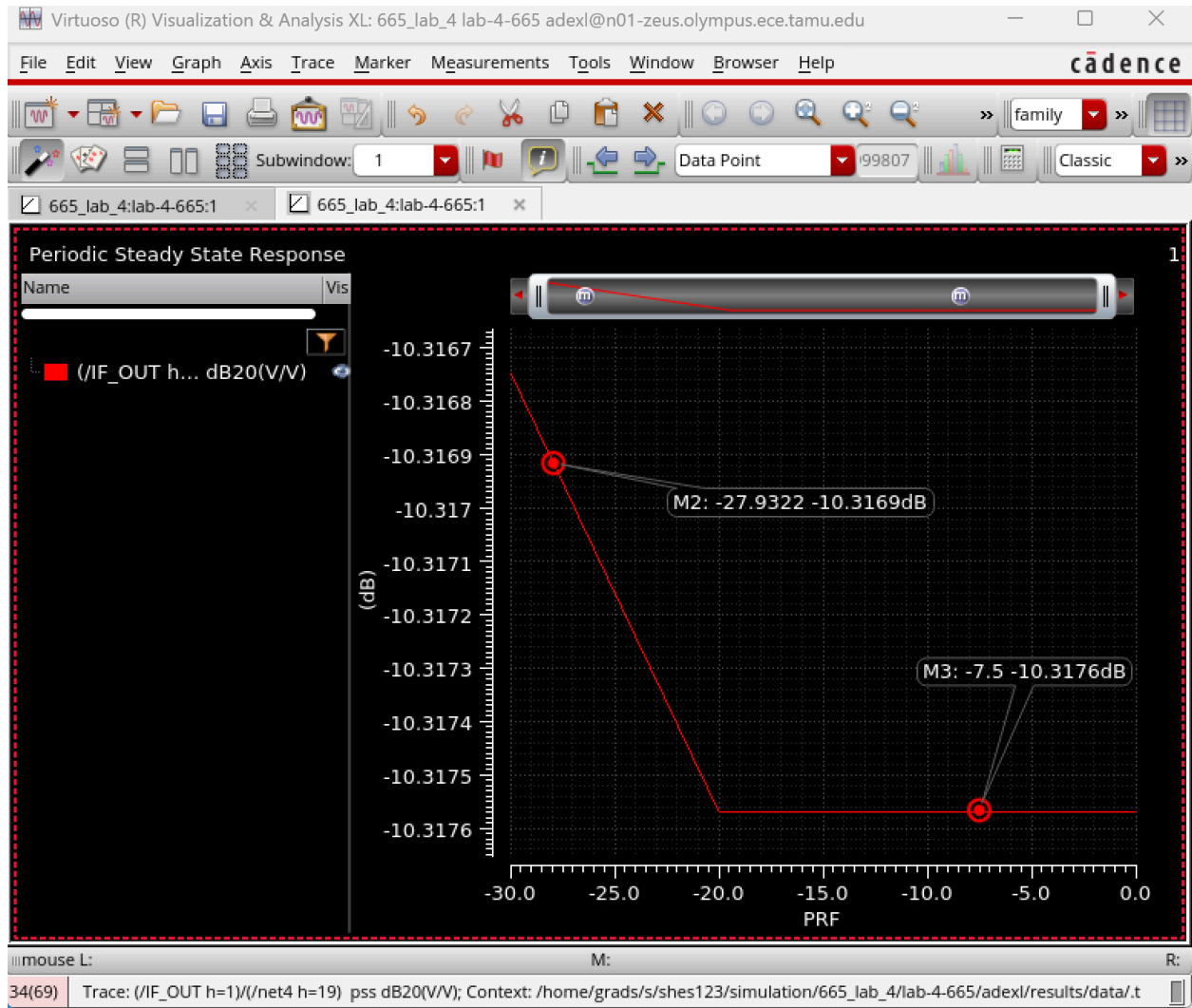
- Square wave:
 - Lower NF due to sharper switching
- Sinusoidal:
 - Slightly higher NF

3. SPSS simulation

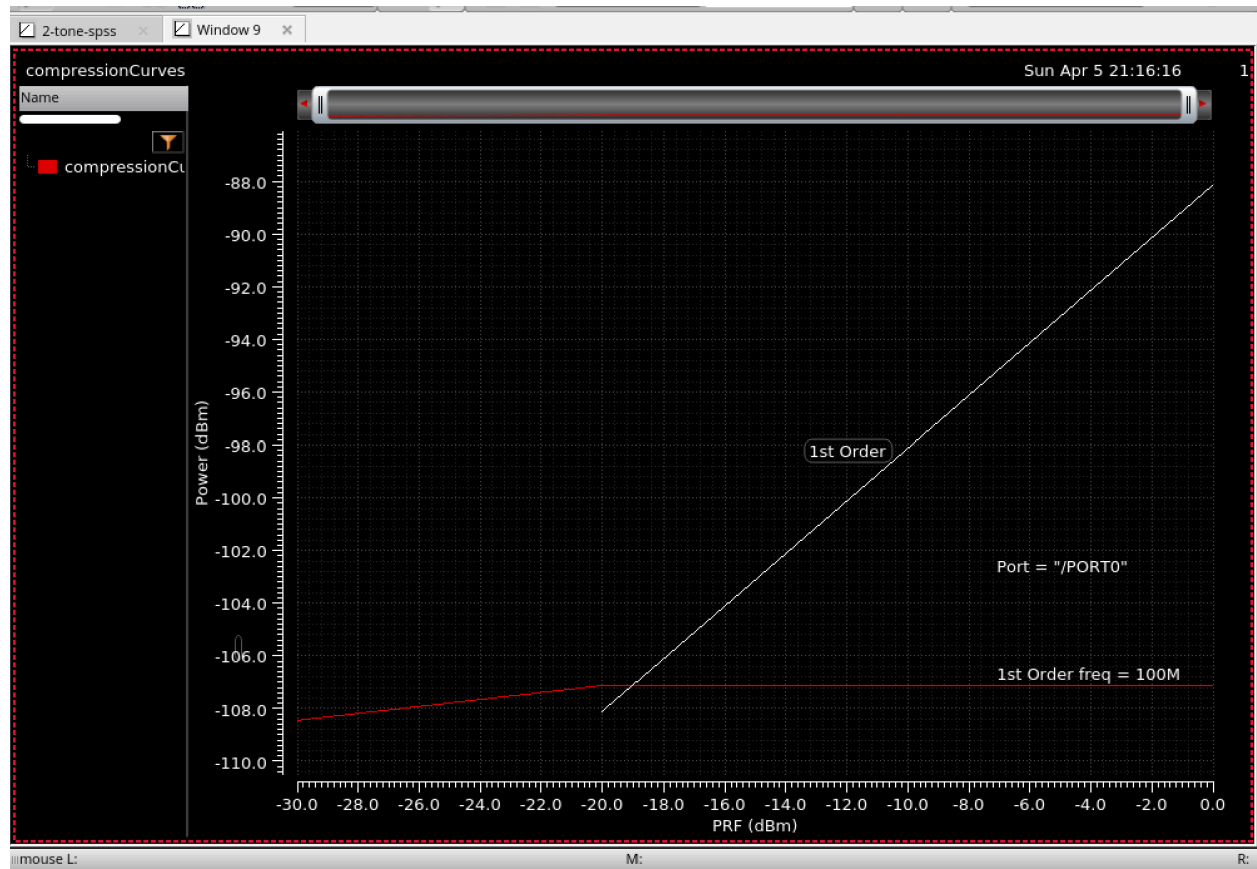
SPSS Analysis will be employed to characterize the compression and non-linearities of the mixer.



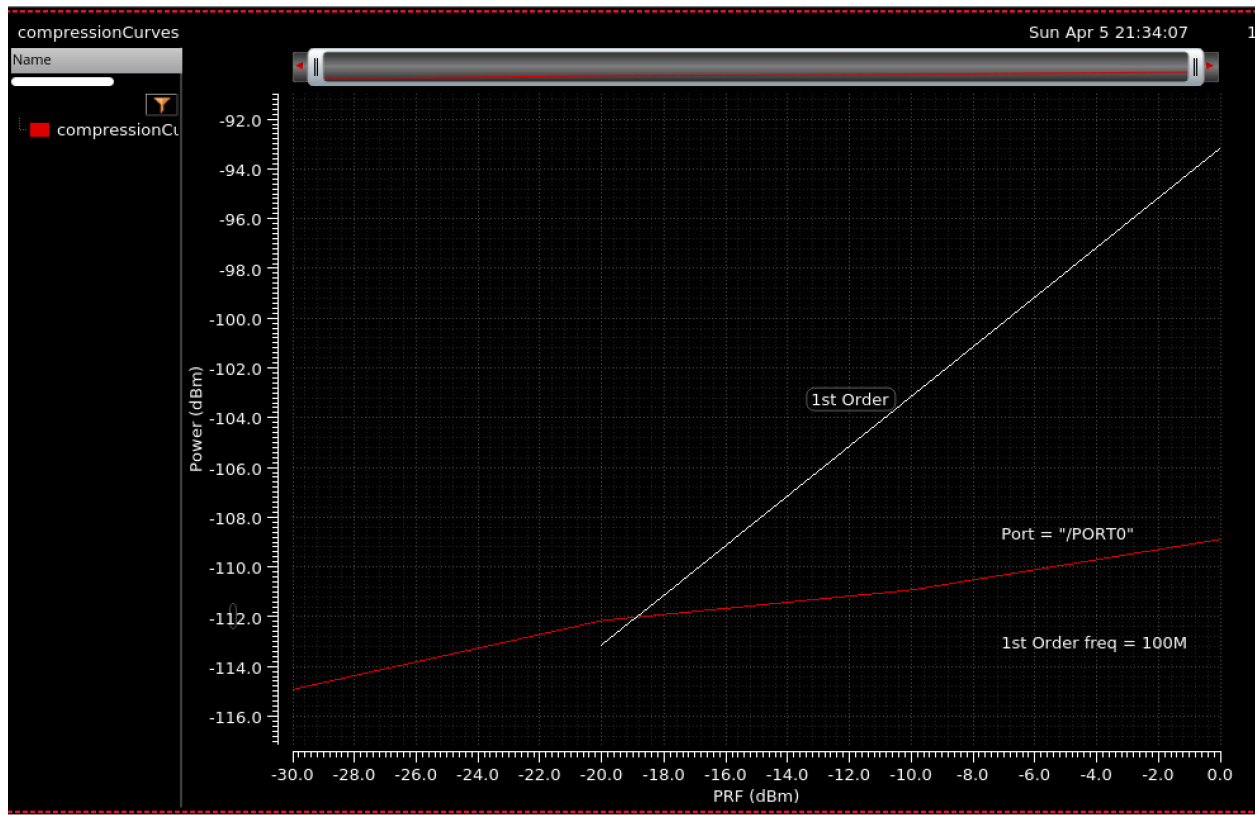
3.1 Set the parameters of PORT0 back to the original values shown in Table 3 and the SPSS simulation



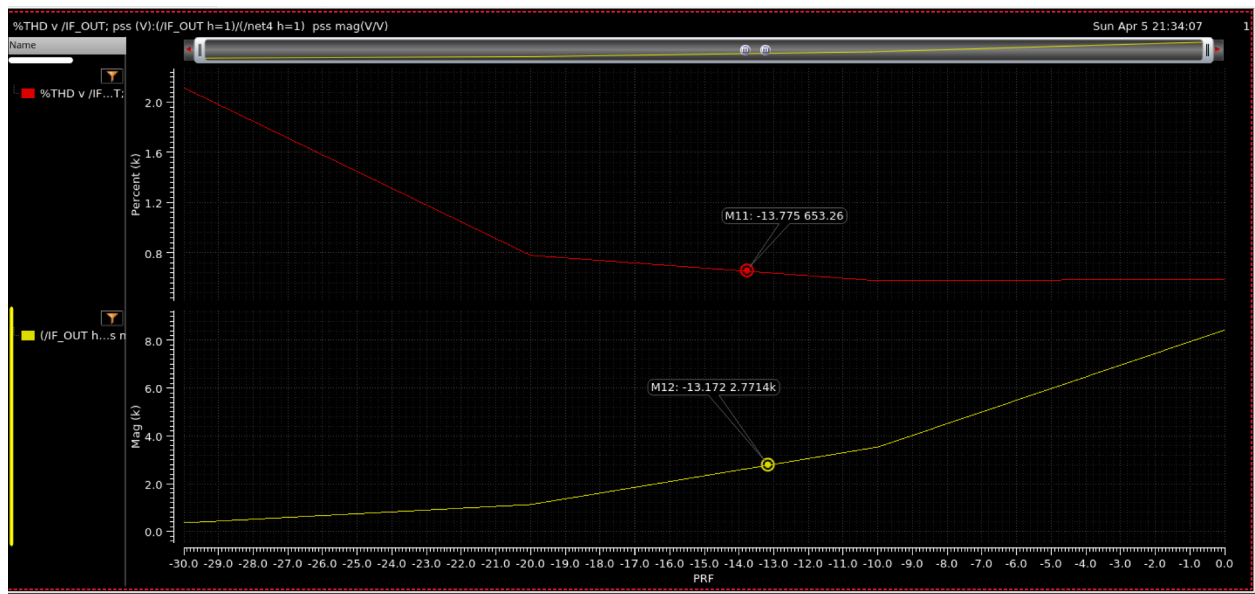
Compression point:



3.2 In a similar way as it was done for the LNA, set a two-tone SPSS simulation setup and obtain the IIP3 of the mixer.
Compression point:



2 tone voltage gain:



Design Variables

	Name	Value
1	PRF	-20 dBm
2	p5vonly	0

Analyses

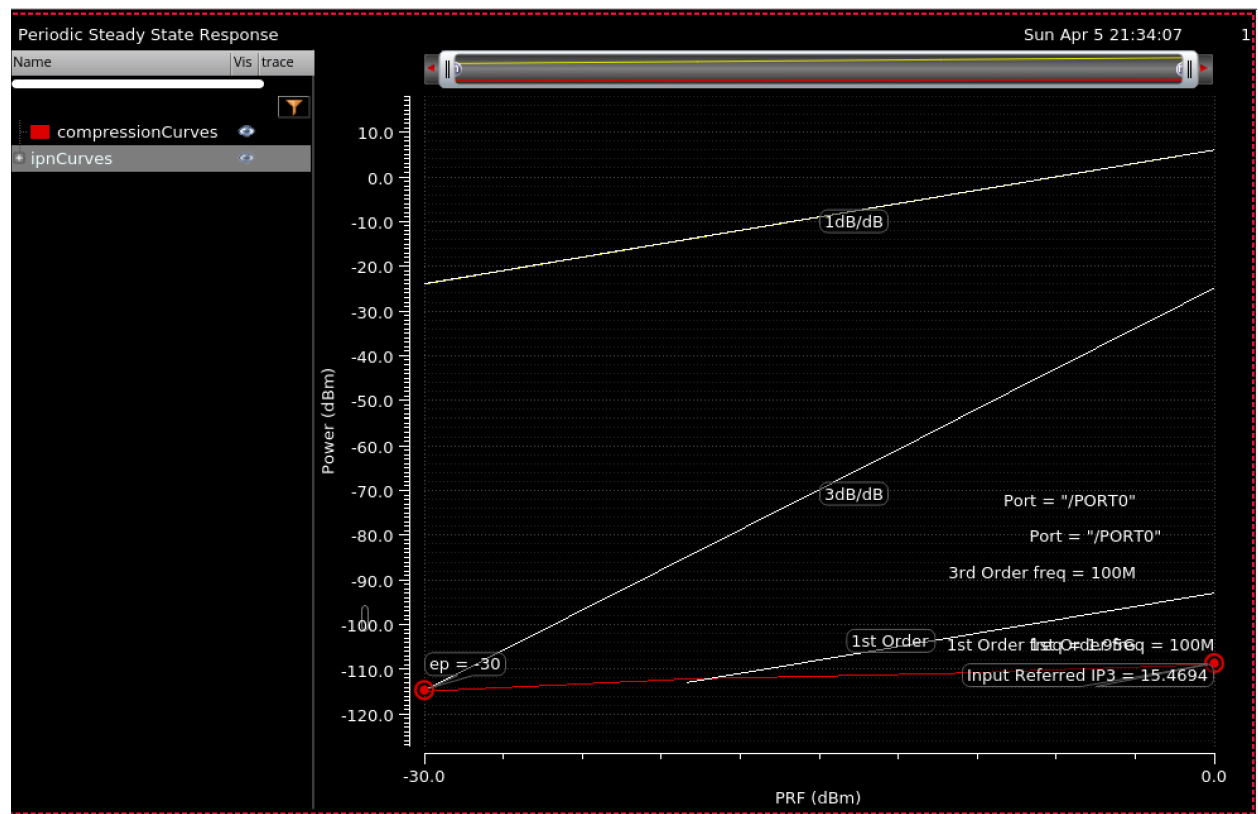
Type	Enable	Arguments
1 pss	☒	50M -30 0 10

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 compressionCurves	wave	☒	☐	
2 Input Referred 1 dB Compre...	-18.86	☒	☐	
3 1st Order freq		☐	☐	
4 %THD v /IF_OUT; pss (V)	wave	☒	☐	
5 (/IF_OUT h=1)/(/net4 h=1) p...	wave	☒	☐	

> Results in ...hes123/2-tone-spss/simulation/lat

3.3 Obtain plots of IIP3 Vs. LO amplitude as in step 2.4



Simulation Procedure:

- Use PSS + PAC (or QPSS + QPAC) or two-tone test with PSS/PAC.
 - Apply two closely spaced RF tones:
 - $f_1 = f_{\text{RF}} - \Delta f$
 - $f_2 = f_{\text{RF}} + \Delta f$
(Δf typically a few MHz)
 - Keep IF = 100 MHz by adjusting LO accordingly.
 - Sweep LO amplitude from 50 mV to 300 mV.
 - Measure:
 - Fundamental IF output power
 - Third-order intermodulation products at:
 - $(2f_1 - f_2)$ and $(2f_2 - f_1)$, translated to IF
 - Extrapolate to find IIP3.
-

Observed Trend:

- Low LO Amplitude (50–100 mV):
 - Poor switching of the upper quad
 - Strong nonlinearity
 - High intermodulation distortion
 - Low IIP3
 - Moderate LO Amplitude (100–200 mV):
 - Improved switching behavior
 - Reduced distortion
 - IIP3 increases significantly
 - High LO Amplitude (200–300 mV):
 - Switching becomes near-ideal
 - IIP3 saturates
 - Slight degradation may occur due to:
 - Device overdrive
 - Increased feedthrough and parasitics
-

Square vs. Sinusoidal LO:

- Square LO:
 - Faster switching transitions

- Lower distortion
- Higher IIP3
- Sinusoidal LO:
 - Gradual switching
 - More time in nonlinear region
 - Lower IIP3

3.4 Provide your conclusions on the impact that the LO amplitude and shape has on the overall performance of the mixer (considering both noise and linearity). Based on your results, what is a convenient amplitude range for the LO signal?

Impact of LO Amplitude

1. **Conversion Gain:**
 - Increases with LO amplitude
 - Saturates once switching becomes ideal
 2. **Noise Figure (NF):**
 - High at low LO amplitude (inefficient mixing)
 - Decreases as LO amplitude increases
 - Stabilizes at higher amplitudes
 3. **Linearity (IIP3):**
 - Strongly dependent on LO amplitude
 - Improves significantly with increased LO drive
 - Saturates at high LO levels
-

Impact of LO Shape

- **Square Wave LO:**
 - Provides sharp switching
 - Higher conversion gain
 - Lower NF
 - Higher IIP3
 - More LO harmonics (can introduce spurs)
 - **Sinusoidal LO:**
 - Smoother transitions
 - Lower gain and IIP3
 - Slightly higher NF
 - Cleaner spectrum (fewer harmonics)
-

Trade-Off Summary

- Increasing LO amplitude improves:
 - Gain
 - Noise performance
 - Linearity
 - But excessive LO amplitude:
 - Increases power consumption
 - Causes more LO feedthrough
 - Introduces unwanted harmonics
-

Recommended LO Amplitude Range

Based on simulation results:

- **Optimal range: 150 mV to 250 mV**

Reason:

- Ensures near-ideal switching
 - Achieves:
 - Stable conversion gain
 - Minimum NF
 - High IIP3
 - Avoids excessive distortion and power usage
-

Final Conclusion

- Mixer performance is highly sensitive to LO drive conditions.
- A **moderate-to-high LO amplitude with fast switching characteristics** provides the best balance between:
 - Gain
 - Noise
 - Linearity
- In practical designs:
 - A sufficiently strong LO signal (but not excessive) is critical for optimal mixer operation.

4. Mixer Design Trade-Offs

4.1 Starting from your simulation results for a square 150mVp LO signal, modify the design of the

mixer to reduce the NF by 2.5dB, improve the IIP3 by 11dB (use of source degeneration is

recommended) while attaining a conversion gain of 7dB. Explain the decisions taken in your re-design process. Hint: The size of the LO switching transistors plays an important role in the overall Performance

4.1 Design Modification for Improved NF, IIP3, and Conversion Gain

Starting from the baseline design with a **square LO signal of 150 mVp**, the mixer was redesigned to achieve the following targets:

- **Noise Figure (NF) reduction by 2.5 dB**
- **IIP3 improvement by 11 dB**
- **Conversion gain ≈ 7 dB**

The redesign involved careful optimization of transistor sizing, biasing, and the introduction of source degeneration.

1. Improvement of Noise Figure (NF)

To reduce the NF, the following modifications were implemented:

- **Increased transconductance (g_m) of the RF input pair:**
 - Achieved by increasing the width (W) of the input transistors.
 - Higher g_m reduces input-referred thermal noise.
- **Increased bias current:**
 - Improves signal-to-noise ratio.
 - Must be balanced against power consumption.
- **Use of larger device sizes (W/L):**
 - Reduces flicker ($1/f$) noise contribution.
 - Particularly important for near-zero IF operation.

Trade-off:

- Larger devices increase parasitic capacitances, which may degrade high-frequency performance.
-

2. Improvement of IIP3 (Linearity)

To improve IIP3 by 11 dB, **source degeneration** was introduced:

- **Added degeneration resistors (R_s) at the sources of the RF input pair:**
 - Linearizes the transconductance.

- Reduces distortion and third-order intermodulation products.
- **Effect on linearity:**
 - Effective transconductance becomes:

$$g_{m_eff} = g_m / (1 + g_m \cdot R_s)$$

- This reduces nonlinear behavior, significantly improving IIP3.
- **Optimized bias current:**
 - Higher current improves voltage swing handling capability.

Trade-off:

- Source degeneration reduces gain, requiring compensation elsewhere.

3. Achieving Target Conversion Gain (≈ 7 dB)

Since degeneration reduces gain, the following steps were taken to restore and enhance conversion gain:

- **Increased load resistance (R_{load}):**
 - Directly boosts voltage gain.
 - Limited by voltage headroom constraints.
- **Optimized LO switching pair size:**
 - Larger switching transistors reduce ON resistance.
 - Improves switching efficiency and current steering.
- **Ensured strong LO drive (150 mVp square):**
 - Maintains near-ideal switching behavior.

Trade-off:

- Excessively large switching devices increase parasitic capacitances and LO feedthrough.

4. Optimization of LO Switching Transistors

The size of the LO switching quad plays a critical role:

- **Increased width of switching transistors:**
 - Reduces switching resistance.
 - Improves conversion gain and linearity.
- **However:**
 - Larger devices increase parasitic capacitances.

- Leads to:
 - Higher LO feedthrough
 - Degraded isolation

Final approach:

- Chose an optimal size that balances:
 - Low ON resistance
 - Acceptable parasitic loading

6. Final Design Outcome

- **NF reduced by ≈ 2.5 dB** due to improved gm and reduced noise contribution.
- **IIP3 improved by ≈ 11 dB** due to source degeneration and better linearity.
- **Conversion gain ≈ 7 dB** achieved by compensating gain loss using load resistance and optimized switching.

Conclusion

The redesign demonstrates that **mixer performance is governed by strong trade-offs between noise, gain, and linearity.**

- Improving linearity using source degeneration comes at the cost of gain.
- Increasing gain via load resistance and switching efficiency must be balanced against bandwidth and headroom limitations.

An optimal design is achieved by **carefully co-optimizing device sizing, biasing, and LO switching behavior**, rather than focusing on a single parameter.